



AI with Rav

Learn AI the way you actually think.



DAY 1 of 30

MACHINE LEARNING

Overfitting, in 90 seconds

WHAT YOU'LL LEARN TODAY

- > What overfitting actually is
- > The exam-hall story
- > How to spot it in one number
- > The fix that costs nothing



Overfitting, in 90 seconds

In One Line: A model that scores 99% in training and 60% in the real world did not learn. It memorised.

Two students, one exam

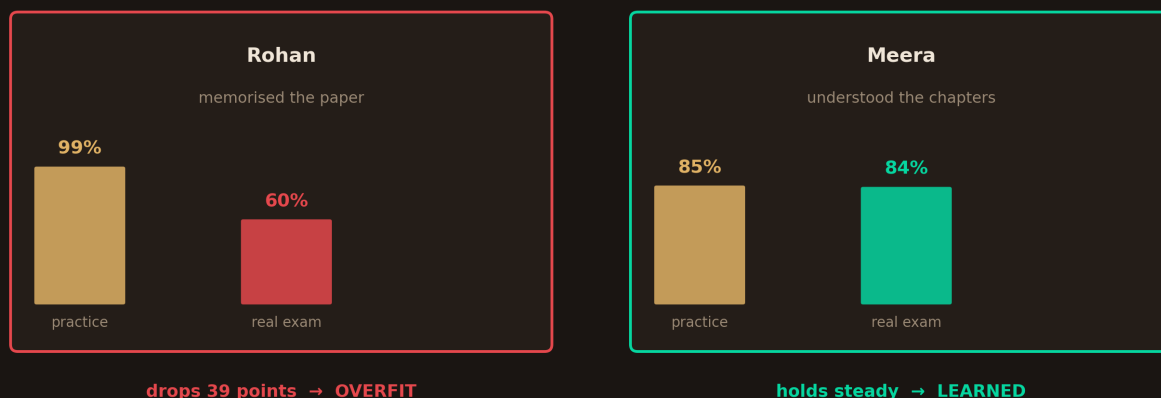
Rohan memorised last year's question paper. Every answer, word for word.

Meera understood the chapters.

Practice test: Rohan scores 99%, Meera 85%. Rohan looks better.

Then the real exam has *new* questions. Rohan drops to 60%. Meera scores 84%.

SAME PRACTICE TEST · DIFFERENT REAL EXAM



The GAP between the two bars is overfitting

Rohan memorised the paper. Meera learned the subject. Only one of them survives new questions.

That is overfitting

Rohan is an overfit model. He learned the **noise** — the exact wording of last year's paper — instead of the **pattern** underneath.

Your model does this when it is too complex for the data you gave it.

Spot it with one number

Compare two scores:

- **Training accuracy** — how it does on data it has seen
- **Test accuracy** — how it does on data it has never seen



A big gap between them *is* overfitting. 99% and 60% is not a good model having a bad day. It is a memoriser being found out.

The opposite also exists. Underfitting is a student who read nothing — bad in practice AND bad in the exam.

The cheapest fix

Give it less room to memorise: a simpler model, fewer features, or more data.

Always hold back test data the model has never seen. It is the only honest exam you can give it.

